Lecture 6: C program structure

- Refer to the **organized way a program** is written, showing its parts and how it runs step by step.

Lesson objectives

By the end of this lesson, the learner will be able to:

- Describe the features of the C language
- Identify the IDEs for C language.
- Describe the steps for creating a C program using a given IDE.
- Describe the structure of a C program.

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- C is a general-purpose, imperative computer programming language, supporting structured programming.
- C was originally developed by Dennis Ritchie between 1969 and 1973 at Bell Labs, and used to re-implement the Unix operating system.

a) Qualities of a good program.

- **Error free** – runs without errors/exceptions
- **Easy to understand** – clear to read and follow
- **Efficient and effective** – uses minimal resources
- **Easy and cheaper to maintain** – simple to update and fix

b) Steps in writing program in C.

- Launch text editor or IDE e.g. Dev C++
- Type source code and save with .c extension.
- Compile the source code to produce object code.
- Build/link the compiled code to produce the executable code.
- Run the executable code.

Structure of a program in C:

i. **Header file** e.g. **#include<stdio.h>**
ii. **Pre-processor directives** e.g. **#include**
iii. **Global variable** declarations e.g. **int a,b**
iv. **Function definitions** e.g**. int Add()**
v. **Main function** e.g. **main()**
vi. **Local variable** declarations
vii. **Printf()** and **scanf()** functions
viii. **Return** statement e.g. **return 0;**

Structure of a program in C: Print string

```
#include<stdio.h>
int main()
{
printf("Hello world\n");
return 0;
}
```

Structure of a program in C: Add two integer numbers

```
#include<stdio.h>
int main()
{
int a,b,c;
printf("Enter first number\n");
scanf("%d",&a);
printf("Enter second number\n");
scanf("%d",&b);
c=a+b;
printf("The result is %d\n",c);
return 0;
}
```

Structure of a program in C: Subtract two integer numbers

```
#include<stdio.h>
int main()
{
int a,b,c;
printf("Enter first number\n");
scanf("%d",&a);
printf("Enter second number\n");
scanf("%d",&b);
c=a-b;
printf("The result is %d\n",c);
return 0;
}
```

Structure of a program in C: Multiply two real numbers

```
#include<stdio.h>
int main()
{
float a,b,c;
printf("Enter first number\n");
scanf("%f",&a);
printf("Enter second number\n");
scanf("%f",&b);
c=a*b;
printf("The result is %.2f\n",c);
return 0;
}
```

Structure of a program in C: Divide two real numbers

```
#include<stdio.h>
int main()
{
float a,b,c;
printf("Enter first number\n");
scanf("%f",&a);
printf("Enter second number\n");
scanf("%f",&b);
c=a/b;
printf("The result is %.2f\n",c);
return 0;
}
```

Structure of a program in C:

```
1   #include <stdio.h>
2   int main()
3   {
4       int a,b,c;
5       printf("Enter first number\n");
6       scanf("%d",&a);
7       printf("Enter second number\n");
8       scanf("%d",&b);
9       c = a * b;
10      printf("The result is %d\n",c);
11      return 0;
12  }
```

- Includes the standard input-output library for using printf() and scanf().
- The main() function is where the program execution begins.
- Beginning of the main function (Opening brace).
- Declare three integer variables a, b and c.
- Display message to ask the user for the first number.
- Read the first number entered by the user and store it in a.
- Display message to ask the user for the second number.
- Read the second number entered by the user and store it in b.
- Multiply a and b, store the result in c.
- Display the result stored in c.
- End the program and return success.
- End of main function (Closing brace).

Structure of a program in C:

	CODE
1	`#include<stdio.h>`
2	`int sum;`
3	`float multiply(int x, float y)`
4	`{`
5	`    return x*y;`
6	`}`
7	`int main(){`
8	`    int a;`
9	`    float b,c;`
10	`    printf("Enter first number\n");`
11	`    scanf("%d",&a);`
12	`    printf("Enter second number\n");`
13	`    scanf("%f",&b);`
14	`    c=multiply(a,b);`
15	`    printf("The result is %f\n",c);`
16	`    return 0;`
17	`}`

	EXPLANATION
1	Includes input-output library (for printf and scanf)
2	Declares a global integer variable 'sum'
3	Starts a function 'multiply' that takes an int (x) and a float (y) and returns float
4	Beginning of function body
5	Returns the product of x and y
6	End of 'multiply' function
7	Start of main function (program begins here)
8	Declares local integer variable 'a'
9	Declares local float variables 'b' and 'c'
10	Displays prompt for first number
11	Reads an integer and stores in 'a'
12	Displays prompt for second number
13	Reads a float and stores in 'b'
14	Calls 'multiply' with a and b; result stored in 'c'
15	Displays the result stored in 'c'
16	Returns 0 (program ended successfully)
17	End of main function

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Uses of main function:

Role of the C main function in programming

a) **Entry point**: program starts execution here.

b) **Program control:** manages flow of instructions and function calls.

c) **Return value:** sends back status (0 = success).

d) **Command-line input:** can receive values when program is run.

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Common mistakes when writing main method:

a) Missing the **return statement** or giving an incorrect return type

b) Use of **incorrect parameter** types or not handling command-line arguments properly

c) Including wrong **header files**

d) Misusing or omitting necessary programming constructs

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Best practices when writing main method

a) Include a return statement with an appropriate return value.

b) Use the correct function signature for your specific needs

c) Use the required header files for standard I/O and other functions

d) Follow proper programming constructs and semantic rules

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Advantages of C

- Highly **portable** language.
- **Modular** structure is easy to debug.
- **Building block** for many languages.
- Easy to code by use of **library functions**.
- Easy to **learn**.

Disadvantages of C

- No support for OOP features
- No support for namespace
- Does not support code reusability.
- Does not support run time type checking.
- Does not support destructors and constructors.

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Program errors: Omission or mistake in the source code

By time detected:

- Compile time errors e.g. syntax and compile
- Run time errors e.g. logical errors and exceptions

By effect on the program:

- Fatal errors e.g. divide by zero
- Non fatal errors e.g. wrong formula

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Example:

```
#include<stdio.h>  //header file
#define PI 3.142     //pre-processor directives
float radius;              //global variables
int main(char args[])  //main function
{
float area, circ;      //local variables
printf("Enter radius of the circle:\n"); //prompt
scanf("%f",&radius); //get values and store in the memory
area=PI*radius*radius; //calculate area
circ=2*PI*radius;  //calculate circumference
printf("The area of the circle is %.4f:\n", area); //display
printf("The circumference of the circle is %.3f:\n",circ);
return 0;  //end
}
```

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Compile &run C program from command prompt

Step 1: change directory

```
C:\Users>CD C:\Users\Desktop
```

Step 2: compile the source file

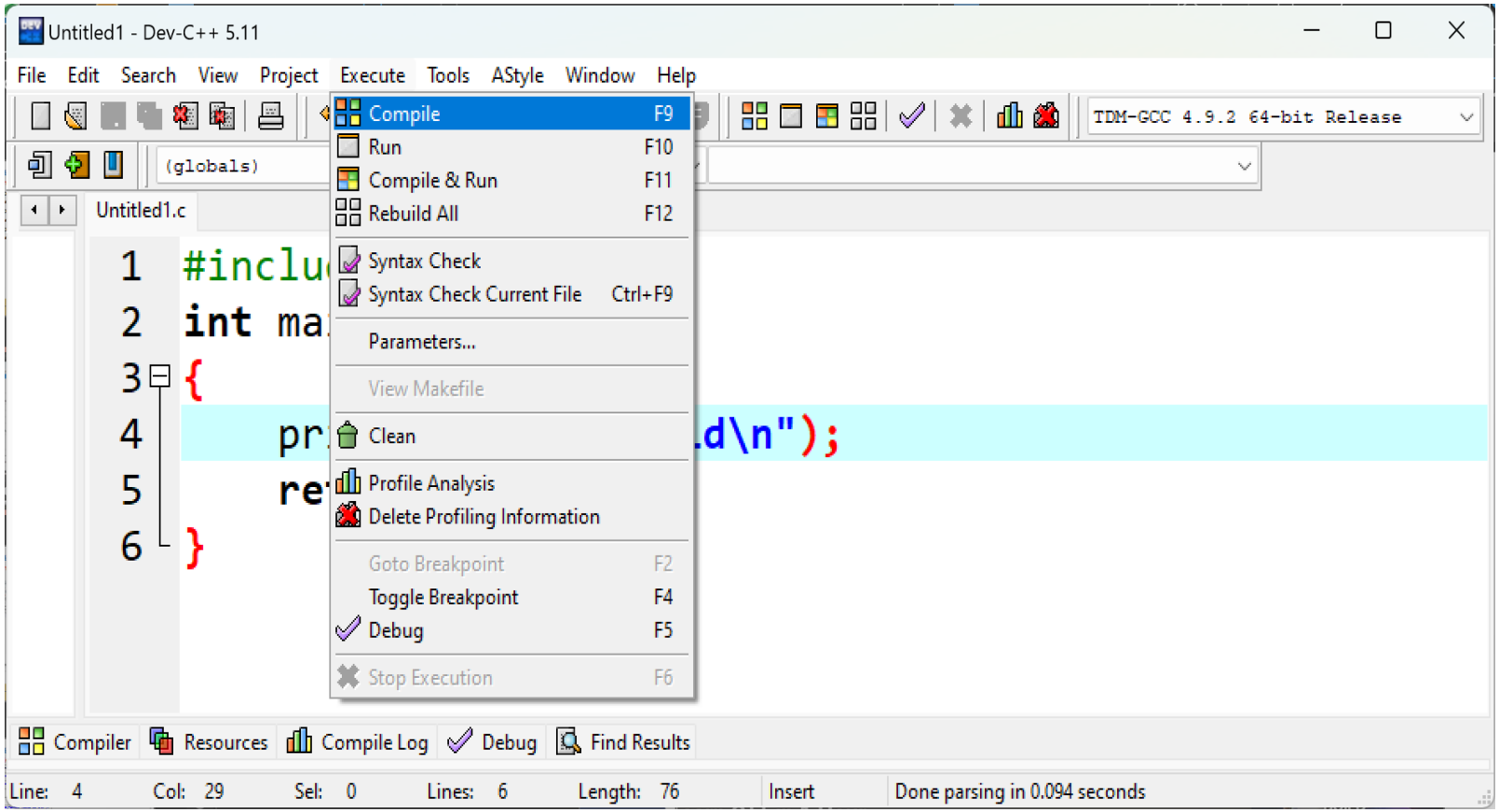
```
C:\Users\Desktop>gcc -o myexe myfile.c
```

Step 3: Run the executable file

```
C:\Users\Desktop>myexe
```

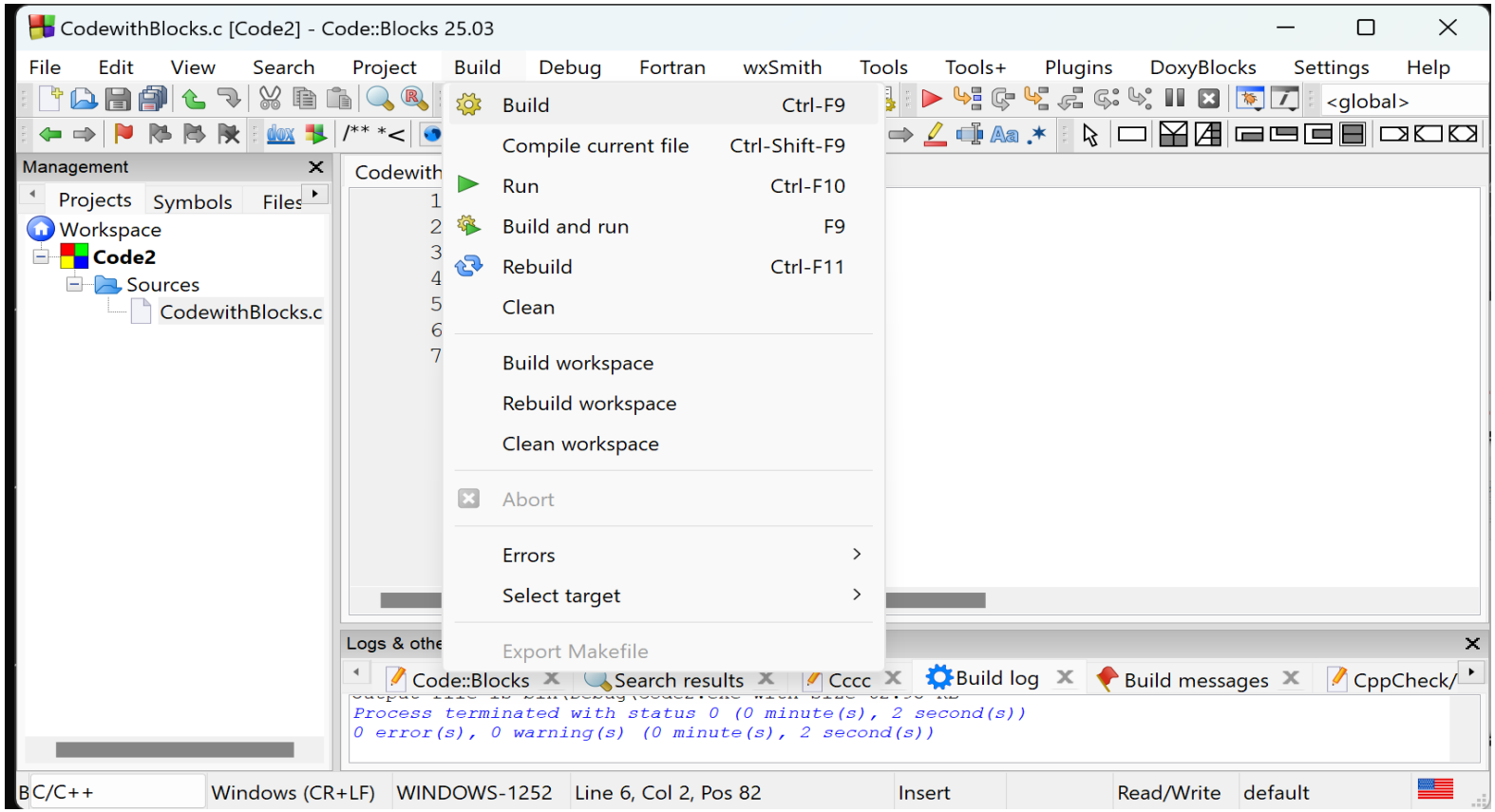
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Compile& run C program from code Blocks IDE



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Compile& run C program from code Blocks IDE



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Activity 6: Complete the table below

LN	Statement	Comment
1	#include<stdio.h>	
2	int main()	//main method begins
3	{	//open bracket for main block
4		//Variables a,b and c to store real numbers
5	printf("enter value of a\n");	
6		//read value of a
7	printf("enter value of b\n");	
8	scanf("%f",&b);	
9		Calculate product of a and b then store in c
10		Display c
11	return 0;	
12	}	

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Exercise 6:

a) Describe any three advantages of C language.

b) Explain what will happen if main function is missing in a C program.

c) Using an example describe the structure of a C program.

d) Define the following terms as used in C language program.

   i. Scanf() function
   ii. Main method
   iii. Pre-processor directive

e) Which error(s) is likely to pass undetected during compilation

   i. Missing semi colon in a statement
   ii. Wrongly spelt identifier
   iii. Wrongly spelt keyword

Reference:

1. An introduction to programming by Wu Thomas, Norman and Theodore: McGrill (1997).
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3. C how to program by H.M. Deitel and P.J.Deitel, 3rd ed: Prentice hall(2001).
4. https://www.w3schools.in/c-tutorial/
5. https://www.tutorialspoint.com/cprogramming/index.htm
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This course has laid foundation for your programming career



THANK YOU